

15 - Expected Cost of Misclassification, The Optimal Classifier, The Bayes Classifier, The MLE Classifier, LDA and QDA

Applied Statistics

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1 The Big Picture — What is Classification?

- **Classification** is the problem of assigning an observed unit to one of g known (or unknown) groups, based on its measured features.
- Each statistical unit is represented by a pair $(\underline{X}^T, L)$ where:
 - $\underline{X} = (x_1, \dots, x_p)^T \in \mathcal{X} \subseteq \mathbb{R}^p$ is the **feature vector**.
 - $L \in \{1, 2, \dots, g\}$ is the **label** (group membership).
- **Goal:** Find a **classifier**, a function

$$\delta : \mathcal{X} \rightarrow \{1, \dots, g\}$$

such that given the features of a unit, δ correctly predicts its label.

- **Two learning situations:**
 - **Supervised (Discriminant Analysis):** Both features $\underline{x}_i$ and labels ℓ_i are observed in the training set. The goal is to learn the optimal δ .
 - **Unsupervised (Cluster Analysis):** Only features $\underline{x}_i$ are observed; labels are hidden. The goal is first to discover the groups, then build δ .

2 The Three Ingredients for a Supervised Classifier

- **Ingredient 1 — Class-conditional densities:**

$$\underline{X} \mid L = i \sim f_i(\underline{x}), \quad f_i : \mathbb{R}^p \rightarrow [0, \infty)$$

- The distribution of features must differ across groups: if $f_i = f_j$, the feature carries no information and classification is impossible.
- *Example:* The number of eyes is the same for males and females — a useless feature!

- **Ingredient 2 — Prior probabilities:**

$$\mathbb{P}[L = i] = p_i, \quad \forall i = 1, \dots, g$$

with requirements: $p_i \geq 0$ and $\sum_{i=1}^g p_i = 1$.

- Priors reflect how common each group is *before* observing any features.
- They come from **context and domain knowledge**, not necessarily from the training data.
- *Example:* At night, the prior probability that a stranger is dangerous is higher than in the morning.

- **Ingredient 3 — Cost of misclassification:**

$c(i | j)$ = cost of predicting group i when the truth is group j

- Requirements: $c(i | j) \geq 0$ for all i, j , and $c(i | i) = 0$ (no cost for correct classification).
- Symmetry is **not** assumed: $c(i | j) \neq c(j | i)$ in general.
- *Example:* Telling a cancer patient they are healthy is far more costly than the reverse.
- Costs come from **problem-domain knowledge**, not from the training set.

3 The Expected Cost of Misclassification (ECM)

- The classifier δ partitions the feature space $\mathcal{X}$ into regions:

$$R_i = \{\underline{x} \in \mathcal{X} : \delta(\underline{x}) = i\} = \delta^{-1}(i), \quad i = 1, \dots, g$$

- $\{R_1, \dots, R_g\}$ is a (finite, measurable) partition of $\mathcal{X}$: $R_i \cap R_j = \emptyset$ for $i \neq j$ and $\bigcup_{i=1}^g R_i = \mathcal{X}$.
- **Specifying δ is equivalent to specifying $\{R_i\}$.**
- **ECM for the binary case ($g = 2$):**

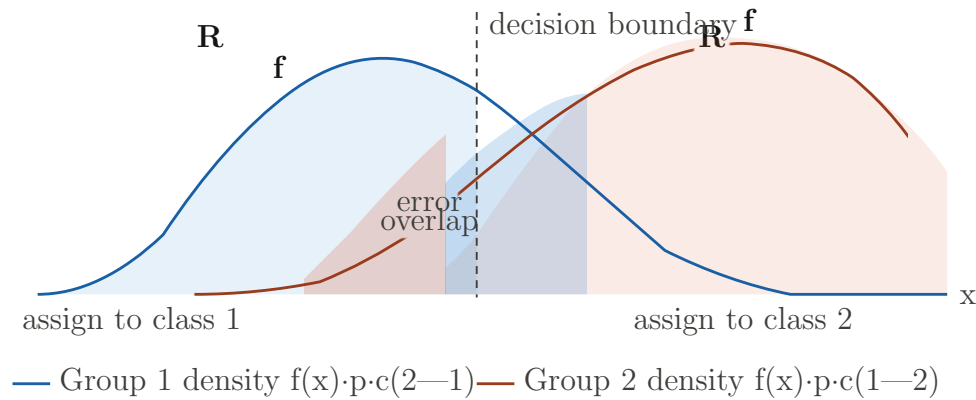
$$ECM(\delta) = \int_{R_2} c(2 | 1) f_1(\underline{x}) p_1 d\underline{x} + \int_{R_1} c(1 | 2) f_2(\underline{x}) p_2 d\underline{x}$$

- First term: cost of misclassifying group-1 units into region R_2 .
- Second term: cost of misclassifying group-2 units into region R_1 .

- **Manipulation of ECM:** Adding and subtracting $\int_{R_1} c(2|1) f_1(\underline{x}) p_1 d\underline{x}$:

$$ECM(\delta) = c(2 | 1) p_1 + \int_{R_1} \underbrace{[c(1 | 2) f_2(\underline{x}) p_2 - c(2 | 1) f_1(\underline{x}) p_1]}_{g(\underline{x})} d\underline{x}$$

- The term $c(2 | 1) p_1$ is a fixed constant — we cannot control it.
- To minimise $ECM(\delta)$, we choose R_1 to be exactly the region where $g(\underline{x}) \leq 0$.



4 The Optimal Classifier Minimising ECM

- **Optimal regions (binary case):**

$$R_1 = \{ \underline{x} \in \mathbb{R}^p : c(1 | 2) f_2(\underline{x}) p_2 \leq c(2 | 1) f_1(\underline{x}) p_1 \}$$

$$R_2 = R_1^c = \{ \underline{x} \in \mathbb{R}^p : c(2 | 1) f_1(\underline{x}) p_1 < c(1 | 2) f_2(\underline{x}) p_2 \}$$

- Assign $\underline{x}$ to class 1 when the weighted evidence for class 1 outweighs that for class 2.
- The boundary shifts with costs and priors — it is **not** simply where $f_1 = f_2$.

- **General case ($g \geq 2$ groups):**

$$ECM(\delta) = \sum_{i=1}^g \int_{R_i} \sum_{k \neq i} c(i | k) f_k(\underline{x}) p_k d\underline{x}$$

- **Optimal regions (general):**

$$R_i = \left\{ \underline{x} \in \mathbb{R}^p : \sum_{k \neq i} c(i | k) f_k(\underline{x}) p_k \leq \sum_{k \neq j} c(j | k) f_k(\underline{x}) p_k, j = 1, \dots, g \right\}$$

- So the optimal classifier is: $\delta(\underline{x}) = i \iff \underline{x} \in R_i$.

- **Equivalent formulation using posterior probabilities:**

Using Bayes' theorem: $\frac{f_k(\underline{x}) p_k}{\sum_{k=1}^g f_k(\underline{x}) p_k} = \mathbb{P}(L = k | \underline{X} = \underline{x})$, the optimal regions become:

$$R_i = \left\{ \underline{x} \in \mathbb{R}^p : \sum_{k \neq i} c(i | k) \mathbb{P}(L = k | \underline{X} = \underline{x}) \leq \sum_{k \neq j} c(j | k) \mathbb{P}(L = k | \underline{X} = \underline{x}), j = 1, \dots, g \right\}$$

- This is the **general Bayesian classifier** — it computes expected posterior costs and assigns to the group with minimum expected cost.
- **Note:** The costs need only be specified up to a multiplicative positive constant — scaling all costs by the same positive number does not change R_i .

5 The Bayes Classifier

- **Assumption:** All misclassification costs are equal:

$$c(i | j) = \text{const} > 0, \quad \forall i \neq j$$

- The constant cancels from both sides of the optimality condition. The optimal regions reduce to:

$$R_i = \{ \underline{x} \in \mathbb{R}^p : \mathbb{P}(L = i | \underline{X} = \underline{x}) \geq \mathbb{P}(L = j | \underline{X} = \underline{x}), j = 1, \dots, g \}$$

- **Rule:** Assign $\underline{x}$ to the group with the **highest posterior probability**.

$$\delta(\underline{x}) =_{i \in \{1, \dots, g\}} \mathbb{P}(L = i | \underline{X} = \underline{x})$$

- **Costs and priors play the same role.** If $c(i | k) = c_k \geq 0$ (cost depends only on the true group), define modified priors:

$$\pi_k = c_k p_k \cdot \frac{1}{\sum_j c_j p_j}, \quad k = 1, \dots, g$$

These satisfy $\pi_k \geq 0$ and $\sum_k \pi_k = 1$. Then:

$$R_i = \left\{ \underline{x} \in \mathbb{R}^p : \sum_{k \neq i} f_k(\underline{x}) \pi_k \leq \sum_{k \neq j} f_k(\underline{x}) \pi_k, j = 1, \dots, g \right\}$$

- This is again a **Bayes classifier**, but with modified priors π_k absorbing the cost structure.
- **Conclusion:** Costs and priors are interchangeable — they both shift the decision boundary.
- **Note:** The Bayes classifier is not always appropriate. If misclassification costs are highly asymmetric (e.g. medical diagnosis), the full optimal classifier with explicit cost structure should be used.

6 The MLE Classifier

- **Assumptions:** Equal costs **and** equal priors:

$$c(i | j) = \text{const} > 0 \quad \text{and} \quad p_1 = \dots = p_g = \frac{1}{g}$$

- Since $\mathbb{P}(L = j | \underline{X} = \underline{x}) = \frac{f_j(\underline{x}) p_j}{\mathbb{P}(\underline{X} = \underline{x})}$ and all priors p_j cancel:

$$R_i = \{ \underline{x} \in \mathbb{R}^p : f_i(\underline{x}) \geq f_j(\underline{x}), j = 1, \dots, g \}$$

- **Rule:** Assign $\underline{x}$ to the group whose density is **largest** at $\underline{x}$:

$$\delta(\underline{x}) =_{i \in \{1, \dots, g\}} f_i(\underline{x})$$

- **Interpretation:** Assign to the group under which the observed features are *most likely* to have occurred — Maximum Likelihood Estimation.
- **Note:** The MLE classifier does not specify costs or priors explicitly; it implicitly assumes they are equal and they cancel. Not every problem can be solved under such assumptions!
- **Where do the ingredients come from?**
 - f_i : estimated from the training data.
 - Priors p_i : from domain knowledge or context, not necessarily the training set.
 - Costs $c(i | j)$: from real-world consequences of errors, not from the training set.

7 Quadratic Discriminant Analysis (QDA)

- **Assumption:** The features in each group follow a **multivariate Gaussian distribution** with group-specific mean and covariance:

$$\underline{X} | L = i \sim \mathcal{N}_p(\underline{\mu}_i, \Sigma_i), \quad i = 1, \dots, g$$

- The density for group i is:

$$f_i(\underline{x}) = \frac{1}{\sqrt{(2\pi)^p \det(\Sigma_i)}} \exp\left(-\frac{1}{2}(\underline{x} - \underline{\mu}_i)^T \Sigma_i^{-1}(\underline{x} - \underline{\mu}_i)\right)$$

- Applying the Bayes classifier rule $f_i(\underline{x}) p_i \geq f_j(\underline{x}) p_j$ and taking the logarithm (monotone transformation, preserves the inequality):

$$\log(p_i) - \frac{1}{2} \log(\det(\Sigma_i)) - \frac{1}{2}(\underline{x} - \underline{\mu}_i)^T \Sigma_i^{-1}(\underline{x} - \underline{\mu}_i) \geq \log(p_j) - \frac{1}{2} \log(\det(\Sigma_j)) - \frac{1}{2}(\underline{x} - \underline{\mu}_j)^T \Sigma_j^{-1}(\underline{x} - \underline{\mu}_j)$$

- **Definition — Quadratic Discriminant Score Functions:**

$$d_i^Q(\underline{x}) = \log(p_i) - \frac{1}{2} \log(\det(\Sigma_i)) - \frac{1}{2}(\underline{x} - \underline{\mu}_i)^T \Sigma_i^{-1}(\underline{x} - \underline{\mu}_i), \quad i = 1, \dots, g$$

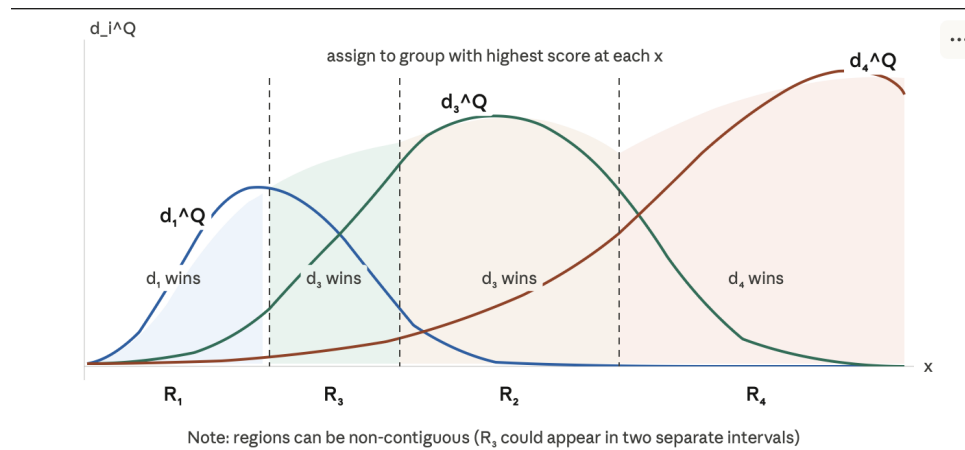
- The three components of $d_i^Q(\underline{x})$:
 - $\log(p_i)$: how common group i is a priori.
 - $-\frac{1}{2} \log(\det(\Sigma_i))$: penalises groups with larger spread.
 - $-\frac{1}{2}(\underline{x} - \underline{\mu}_i)^T \Sigma_i^{-1}(\underline{x} - \underline{\mu}_i)$: the **squared Mahalanobis distance** from $\underline{x}$ to the mean $\underline{\mu}_i$, adjusted for the shape of group i .

- **QDA classifier:**

$$\delta(\underline{x}) = i \iff d_i^Q(\underline{x}) \geq d_j^Q(\underline{x}), \quad \forall j = 1, \dots, g$$

$$R_i = \{\underline{x} \in \mathbb{R}^p : d_i^Q(\underline{x}) \geq d_j^Q(\underline{x}), \quad j = 1, \dots, g\}$$

- **Why “quadratic”?** The term $(\underline{x} - \underline{\mu}_i)^T \Sigma_i^{-1}(\underline{x} - \underline{\mu}_i)$ is quadratic in $\underline{x}$ (it contains $\underline{x}^T A \underline{x}$ terms). Since each group has a *different* Σ_i , these quadratic terms do not cancel when comparing groups, producing **curved decision boundaries**.
- **Note:** QDA requires estimating $\underline{\mu}_i$ and Σ_i for each group — it needs sufficient data per group. The boundaries can be ellipses, parabolas, or hyperbolas.



8 Linear Discriminant Analysis (LDA)

- **Additional assumption:** All groups share the **same covariance matrix**:

$$\Sigma_1 = \Sigma_2 = \dots = \Sigma_g = \Sigma$$

- Since Σ is shared, $\det(\Sigma)$ is the same for all groups and cancels. Expand the quadratic term:

$$\frac{1}{2}(\underline{x} - \underline{\mu}_i)^T \Sigma^{-1} (\underline{x} - \underline{\mu}_i) = -\frac{1}{2} \underline{x}^T \Sigma^{-1} \underline{x} + \underline{\mu}_i^T \Sigma^{-1} \underline{x} - \frac{1}{2} \underline{\mu}_i^T \Sigma^{-1} \underline{\mu}_i$$

- The term $-\frac{1}{2} \underline{x}^T \Sigma^{-1} \underline{x}$ is **identical** for all groups and cancels when comparing $d_i^Q \geq d_j^Q$.
- **Definition — Linear Discriminant Score Functions:**

$$d_i(\underline{x}) = \log(p_i) + \underline{\mu}_i^T \Sigma^{-1} \underline{x} - \frac{1}{2} \underline{\mu}_i^T \Sigma^{-1} \underline{\mu}_i, \quad i = 1, \dots, g$$

These are **linear in $\underline{x}$** (the dot product $\underline{\mu}_i^T \Sigma^{-1} \underline{x}$ is a linear function of $\underline{x}$).

- **LDA classifier:**

$$\delta(\underline{x}) = i \iff d_i(\underline{x}) \geq d_j(\underline{x}), \quad \forall j = 1, \dots, g$$

$$R_i = \{\underline{x} \in \mathbb{R}^p : d_i(\underline{x}) \geq d_j(\underline{x}), \quad j = 1, \dots, g\}$$

- **Why “linear”?** The quadratic term $\underline{x}^T \Sigma^{-1} \underline{x}$ cancels (shared Σ). What remains is linear in $\underline{x}$. The boundary $\{d_i(\underline{x}) = d_j(\underline{x})\}$ is a **hyperplane** (a straight line in 2D).
- **Note:** LDA is very robust — it works well even when the data are not perfectly Gaussian, and it requires far less data than QDA since only one Σ is estimated.

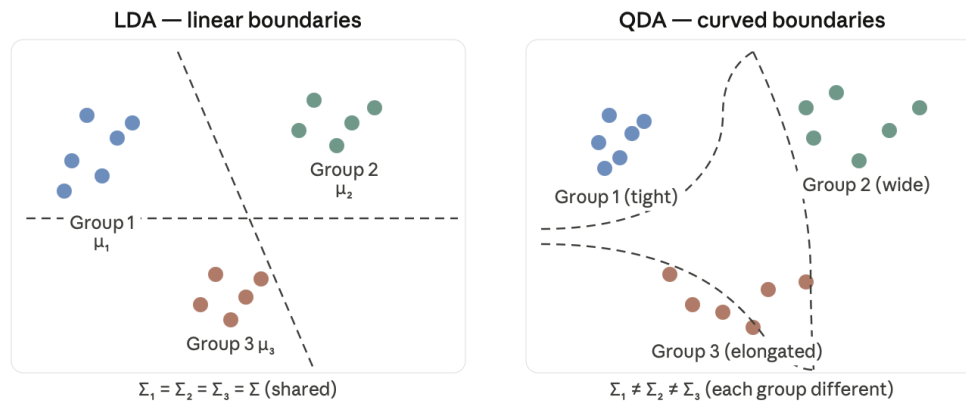


Figure 1: Enter Caption

9 QDA vs. LDA — Summary and Comparison

- **QDA** assumes each group has its own Σ_i (different shape):
 - More flexible, fits the data better when shapes genuinely differ.
 - Requires estimating $\underline{\mu}_i$ and Σ_i per group — needs more data.
 - Decision boundaries are **curved** (quadratic).
 - Weakens without truly Gaussian data.
- **LDA** assumes all groups share Σ (same shape, different centres):
 - Simpler, more robust, needs less data.
 - Decision boundaries are **flat** (linear hyperplanes).
 - Works well even when Gaussian assumption is only approximate.
 - Can be derived without assuming Gaussianity at all (next lecture).
- **Hierarchy of classifiers:**
 - Optimal (ECM) $\supset$ Bayes (c equal) $\supset$ MLE (c and p equal)
 - QDA (Gaussian, Σ_i different) $\supset$ LDA (Gaussian, Σ shared)
- **Key intuition:** LDA asks “Which mean is $\underline{x}$ closest to (in the Mahalanobis sense with shared Σ)?” QDA asks the same but uses each group’s own metric.

Türkçe Açıklamalar / Turkish Explanations

1. Büyük Resim — Sınıflandırma Nedir?

- **Sınıflandırma:** Bir birimi, ölçülen özelliklerine bakarak g gruptan birine atamak.
- Her birim $(\underline{X}^T, L)$ çiftiyle temsil edilir:
 - $\underline{X} = (x_1, \dots, x_p)^T$: **özellik vektörü** (ağırlık, renk, vb.)
 - $L \in \{1, \dots, g\}$: **etiket** (hangi gruba ait?)
- **Amaç:** $\delta(\underline{X})$ = tahmin edilen grup olan bir fonksiyon bulmak.
- **İki durum:**
 - **Gözetimli (Discriminant Analysis):** Hem özellikler hem etiketler bilinir.
 - **Gözetimsiz (Cluster Analysis):** Etiketler gizlidir; önce gruplar keşfedilir.

2. Üç Temel Bileşen

- **Bileşen 1 — Koşullu yoğunluklar:** $\underline{X} \mid L = i \sim f_i(\underline{x})$
 - Grupların özellik dağılımları farklı olmalıdır; yoksa sınıflandırma imkânsızdır.
 - *Örnek:* Gözlerin sayısı erkek ve kadınlarda aynı — işe yaramaz bir özellik!
- **Bileşen 2 — Önceki olasılıklar (Prior):** $\mathbb{P}[L = i] = p_i$
 - Herhangi bir özellik gözlemlenmeden önce bir birimin hangi grupta olduğunun olasılığı.
 - Bağlam ve alan bilgisinden gelir, eğitim verisinden değil.
 - *Örnek:* Gece bir yabancıнын tehlikeli olma ihtimali gündüzden yüksektir.
- **Bileşen 3 — Yanlış sınıflandırma maliyeti:** $c(i \mid j)$
 - Gerçek grup j iken, i tahmin etmenin bedeli.
 - $c(i \mid i) = 0$ (doğru tahmin bedelsiz), simetri gerekmez.
 - *Örnek:* Kanserli bir hastayı sağlıklı demek, tersinden çok daha tehlikeli.

3. Beklenen Yanlış Sınıflandırma Maliyeti (ECM)

- δ sınıflandırıcısı, özellik uzayını $R_1, \dots, R_g$ bölgelerine böler.
- **İki grup için ECM:**

$$ECM(\delta) = \int_{R_2} c(2 \mid 1) f_1(\underline{x}) p_1 d\underline{x} + \int_{R_1} c(1 \mid 2) f_2(\underline{x}) p_2 d\underline{x}$$

- 1. terim: Grup 1'dekileri yanlışlıkla R_2 'ye atama maliyeti.
- 2. terim: Grup 2'dekileri yanlışlıkla R_1 'e atama maliyeti.
- Dönüşüm yapılıncı:

$$ECM(\delta) = c(2 \mid 1) p_1 + \int_{R_1} [c(1 \mid 2) f_2(\underline{x}) p_2 - c(2 \mid 1) f_1(\underline{x}) p_1] d\underline{x}$$

- $c(2|1)p_1$ sabittir, değiştiremeyiz. Sadece R_1 'i seçerek integralin altındaki terimi minimize ederiz.
- **Amaç:** $g(\underline{x}) \leq 0$ olan bölgeyi R_1 olarak seç.

4. ECM'yi Minimize Eden Optimal Sınıflandırıcı

- **İki grup için optimal bölgeler:**

$$R_1 = \{\underline{x} : c(1|2)f_2(\underline{x})p_2 \leq c(2|1)f_1(\underline{x})p_1\}$$

- Grup 1'in ağırlıklı kanıtı $(c(2|1)f_1p_1)$ Grup 2'ninkinden $(c(1|2)f_2p_2)$ büyükse, Grup 1 de.
- Sınır, maliyet ve prior'a göre kayar — sadece $f_1 = f_2$ noktasında değildir.

- **Genel durum ($g \geq 2$):**

$$R_i = \left\{ \underline{x} \in \mathbb{R}^p : \sum_{k \neq i} c(i|k) \mathbb{P}(L = k | \underline{X} = \underline{x}) \leq \sum_{k \neq j} c(j|k) \mathbb{P}(L = k | \underline{X} = \underline{x}), j = 1, \dots, g \right\}$$

- Posterior olasılıkları hesapla, beklenen maliyeti en düşük olan grubu seç.
- **Not:** Maliyetler pozitif sabit katına kadar belirsizdir — hepsini aynı sabit ile çarpmak R_i 'yi değiştirmez.

5. Bayes Sınıflandırıcısı

- **Varsayım:** Tüm yanlış sınıflandırma maliyetleri eşit ($c(i|j) = \text{sabit} > 0$).
- Sabit silinir, optimal bölgeler şöyle sadeleşir:

$$R_i = \{\underline{x} : \mathbb{P}(L = i | \underline{X} = \underline{x}) \geq \mathbb{P}(L = j | \underline{X} = \underline{x}), j = 1, \dots, g\}$$

- **Kural:** En yüksek posterior olasılığa sahip grubu seç.
- **Maliyet ve prior'lar aynı rolü oynar:** $c(i|k) = c_k$ formunda olursa, maliyeti prior'a katla:

$$\pi_k = \frac{c_k p_k}{\sum_j c_j p_j}$$

Ve yeni prior π_k ile standart Bayes sınıflandırıcısı kullan. **Sonuç:** İkisi aynı düğmeyi çeviriyor.

- **Örnek:** Bir e-postanın spam mi yoksa normal mi olduğunu sınıflandırıyorsun. Posterior olasılık spam için %85 ise $\rightarrow$ Spam de.

6. MLE Sınıflandırıcısı

- **Varsayım:** Maliyetler eşit ve prior'lar da eşit ($p_1 = \dots = p_g = 1/g$).
- Her şey silinir; sadece yoğunluk fonksiyonu kalır:

$$\delta(\underline{x}) =_i f_i(\underline{x})$$

- **Kural:** Gözlemlenen $\underline{x}$ için yoğunluğu en yüksek olan grubu seç.
- **Yorum:** “Bu özellikler hangi grubun tipik üyesine en çok benziyor?”
- **Örnek:** Bir kişi 2 metre boyunda. Bu özellik erkekler için çok tipik, kadınlar için çok nadir. MLE $\rightarrow$ Erkek.
- **Not:** MLE her problem için uygun değildir — asimetrik maliyetler varsa kullanma!

7. QDA — Quadratic Discriminant Analysis

- **Varsayım:** Her grubun özellikleri farklı parametrelili Gaussian dağılım izler:

$$\underline{X} \mid L = i \sim \mathcal{N}_p(\underline{\mu}_i, \Sigma_i)$$

- **QDA Skor Fonksiyonu:**

$$d_i^Q(\underline{x}) = \log(p_i) - \frac{1}{2} \log(\det(\Sigma_i)) - \frac{1}{2}(\underline{x} - \underline{\mu}_i)^T \Sigma_i^{-1}(\underline{x} - \underline{\mu}_i)$$

- Üç parça:
 - $\log(p_i)$: Bu grubun ne kadar yaygın olduğu.
 - $-\frac{1}{2} \log(\det(\Sigma_i))$: Bu grubun ne kadar geniş yayıldığı.
 - $-\frac{1}{2}(\underline{x} - \underline{\mu}_i)^T \Sigma_i^{-1}(\underline{x} - \underline{\mu}_i)$: Noktanın bu grubun merkezine Mahalanobis uzaklığı.
- **Kural:** En yüksek skoru veren grubu seç: $\delta(\underline{x}) = \underset{i}{\operatorname{argmax}} d_i^Q(\underline{x})$.
- **Neden “kuadratik”?** Her grubun farklı Σ_i ’si var, kuadratik terimler silinmez $\rightarrow$ **kavisli sınırlar**.
- **Örnek:** Elmalar küçük-yuvarlak (dar dağılım), portakallar büyük-değişken (geniş dağılım). QDA her birinin şeklini öğrenir.

8. LDA — Linear Discriminant Analysis

- **Ek varsayım:** Tüm grupların kovaryans matrisi aynı: $\Sigma_1 = \dots = \Sigma_g = \Sigma$.
- Kuadratik terim $\frac{1}{2} \underline{x}^T \Sigma^{-1} \underline{x}$ her grupta aynı olduğu için **silinir**.
- **LDA Skor Fonksiyonu (doğrusal):**

$$d_i(\underline{x}) = \log(p_i) + \underline{\mu}_i^T \Sigma^{-1} \underline{x} - \frac{1}{2} \underline{\mu}_i^T \Sigma^{-1} \underline{\mu}_i$$

- **Kural:** En yüksek skoru veren grubu seç: $\delta(\underline{x}) = \underset{i}{\operatorname{argmax}} d_i(\underline{x})$.
- **Neden “doğrusal”?** $\underline{x}$ sadece birinci kuvvetiyle geçiyor $\rightarrow$ **düz (hyperplane) sınırlar**.
- **Örnek:** Elmalar ve portakallar benzer şekilde dağılmış ama farklı ağırlıklarda. LDA aralarına düz çizgi çizer.
- **Not:** LDA daha az veri gerektirir, Gaussian olmayan durumlarda bile iyi çalışır.

9. QDA ve LDA Karşılaştırması

- **QDA:**

- Her grup kendi şekline (Σ_i) sahip.
- Daha esnek, daha fazla veri gerektirir.
- Sınırlar: **kavisli** (eğri, elips, parabol).

- **LDA:**

- Tüm gruplar aynı şekle (Σ) sahip, sadece merkezler farklı.
- Daha basit, daha sağlam, daha az veri yeterli.
- Sınırlar: **düz** (doğrusal, hyperplane).

- **Hiyerarşi:**

- Optimal $\supset$ Bayes $\supset$ MLE
- QDA $\supset$ LDA

- **Temel sezgi:** LDA “ x hangi ortalamaya daha yakın?” diye sorar. QDA aynı soruyu sorar ama her grubun kendi metriğiyle.